

Problem Statements

1. Smart Parking & Curbside Availability Predictor

Problem Statement: Build a system that predicts real-time/near-term parking availability in a given area using historical patterns, time, events, and weather, and recommends the best nearby spot before the driver arrives.

- Uses historical/simulated occupancy data to train a prediction model
- Output: map-based UI showing predicted availability
- Stretch: dynamic pricing suggestion, live traffic integration
- Tests: data handling, basic ML/heuristics, UI presentation

2. Driver Behavior & Safety Scorecard

Problem Statement: Using phone sensor or trip-log data, build an app/dashboard that scores a driving trip on harsh braking, sharp turns, over-speeding, and idle time, and gives a gamified safety score with tips.

- Detects driving events from accelerometer/GPS data
- Converts raw signals into a composite safety score
- Stretch: insurance-style risk tiering, peer comparison, weekly trends
- Tests: signal processing, mobile/sensor data handling, product framing

3. EV Route Planner with Charging-Stop Optimization

Problem Statement: Build a route planner for EV drivers that factors in battery range, charging station locations, and charging speed to suggest the optimal route with minimal total trip time.

- Computes feasible route + charging stops from start/end + battery %
- Uses static/mock charging station data
- Stretch: elevation/weather impact on range, real-time occupancy, cost optimization
- Tests: algorithmic/graph thinking, optimization skills

4. Gamified Eco-Driving & Carbon Footprint Coach

Problem Statement: Build an app that analyzes a driver's trip data (speed, acceleration, idle time, route) to score fuel/energy efficiency, estimate CO2 emissions, and gamify improvement with challenges and a leaderboard.

- Converts trip data into an efficiency + emissions score
- Gamification layer: streaks, challenges, peer leaderboard
- Stretch: personalized eco-tips, EV vs. ICE comparison mode, weekly reports
- Tests: data analytics, product/UX thinking, sustainability angle (good talking point for judges)

5. Ride-Hailing Demand Heatmap Predictor

Problem Statement: Build a tool for ride-hailing/taxi drivers that predicts demand hotspots by time and location using historical trip data, recommending where a driver should position themselves next.

- Uses historical/simulated trip-request data to train a prediction model

- Output: map-based heatmap of predicted demand by time window
- Stretch: driver earnings simulation, multi-driver load balancing
- Tests: data handling, basic ML/heuristics, map-based UI presentation